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Project Title:	Spectral Imaging within The EDEN ISS Project – An Antarctic Analog for Enhancing Exploration Life Support		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (Co PI: University of Florida)		
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Task Description:

This project brings established research from NASA funded science on the International Space Station (ISS) into an international project led by DLR (German Space Agency) toward the functionalization of plant growth in support of human exploration. Imaging technologies developed as part of ongoing ISS science will be deployed within the Antarctic EDEN ISS plant production unit to provide tele-science support and plant science analytics to better understand the responses of plants, and production support to optimize plant performance.
Plant production for advanced life support within deep space and extraterrestrial colonies has long been an appreciated area of technology development. Indeed ISS science is now turning toward small scale plant production in space, and plant production units in extreme environments on Earth have proven the value of fresh food. However, there has as yet to be any integrated tests of plant production within space vehicle-like structures in extreme analog environments. To address this gap in knowledge, the EDEN ISS project by the DLR German space agency is designed to place an ISS-inspired module in the context of an established Antarctic station, drawing together for the first time structures and production units designed to space vehicle formats together with the challenges of an extreme and isolated environment. In keeping with the spirit and intent of the EDEN ISS project, the current proposal seeks to fund US participation, bringing plant imaging technologies developed, in part from extensive Shuttle and ISS plant growth experiments, into the science to be conducted within EDEN ISS. The area of fidelity to be addressed by this project is a series of science investigations designed to further planetary research in terrestrial extreme environments that may be analogous to those found on other planets.

The EDEN ISS project is constructing essential controlled environment agriculture technologies using both an International Standard Payload Rack cultivation system to relate to on-board ISS and cis-lunar deployments for crew food supplementation, and a Future Exploration Greenhouse that is designed with respect to future planetary bioregenerative life support system deployments. The overall physical facility, referred to as the EDEN ISS Mobile Test Facility, will provide realistic mass flow and will deploy to the highly-isolated Antarctic Neumayer Station III, operated by the Alfred Wegener Institute.

Among the technologies to be developed and deployed will be imagers designed by the University of Florida (UF) Principal Investigator (PI) team, through this proposed project, to monitor plant health within the Mobile Test Facility. Our major goal is to equip the plant growth systems with information-rich imaging such that plant health and basic science data can be reviewed by a large and integrated science backroom back on “Earth.” An additional goal is to optimize the conceptual connections between EDEN ISS and current ISS capabilities and operations. These two goals bring a directed focus to the project, connecting the planetary surface analog concepts to near term operations of ISS and potential cis-lunar habitats.

To accomplish these goals, spectral imaging of plants will be incorporated into typical data monitoring systems and operation of the project. EDEN ISS, as a large multinational project, has an extensive Data Management Plan (DMP). The work proposed here will depend and draw upon the EDEN ISS DMP for the overall project data, and will specifically draw Mobile Test Facility (MTF) plant growth images and operations images as primary data for analysis and management by the UF team. The UF team will archive and make available all plant health images specific to our tasks via a separate project within the NASA Life Sciences Data Archive (LSDA). The technology is based on small, robust cameras that image wavelengths related to previous and future ISS hardware such that data from EDEN ISS will provide feed forward science for known and anticipated ISS and habitat plant growth hardware.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Exploration mission scenarios are increasingly engaging plant production as adjunct or major parts of life support. This is especially true for deep space and extraterrestrial surface scenarios, and there have been several lower fidelity projects that include plant production in analog tests of such scenarios. Plant growth has been a part of some simulations, and small-scale production type plant growth units are currently on the ISS. However, in none of these cases has an exploration-realistic plant growth facility been incorporated into a challenging analog site such as Antarctica to move the technology significantly toward anticipated mission structures and operations. And none incorporated sophisticated plant imaging into remote science backrooms. The EDEN ISS project is envisioned to fill that large gap, placing an ISS-inspired module near the German Antarctic Neumayer Station III. The EDEN ISS project is an active, funded project that involves a wide international team and is led by the German Aerospace Center, DLR. We have been invited to contribute our presence, expertise, and imaging technology to EDEN ISS. Opportunities for non-European participation at Neumayer III are rare, and EDEN ISS is unique in its magnitude and fidelity. This proposal seeks to seize the opportunity for important imaging technology advancement and US participation within the multinational context of the EDEN ISS project.
The EDEN ISS project seeks to create a structurally and operationally realistic analog to a space station, vehicular, or planetary habitat plant growth module for life support. As such, the project falls under the purview of Space Exploration and Spaceport Technology Development (SESTD) in several subcategories: In-Situ Resource Utilization Technologies (in that EDEN is dedicated to local production of food for bioregenerative Advanced Life Support), Lunar and Mars Base/Settlement Technologies (in that we will be within a robust simulation of life support technologies), and also Advanced Sensors and Detectors (in that we will be developing imaging cameras for monitoring plant health within the facility). As a working production facility, EDEN ISS will provide fresh produce to Neumayer III overwintering crews. And in the process, the project will validate key plant growth system technologies and operational protocols for use on the ISS and within future planetary bioregenerative life support systems -- a key set of technologies for long duration spaceflight and planetary habitation.

The project also has the potential to substantially contribute to plant health monitoring technology in terrestrial environments. Protected agriculture (production greenhouses and other crop facilities that are environmentally controlled or where climate is ameliorated) is a rapidly growing field in a world with expanding needs for food, and expanding regions of marginal crop land. Robotic tending and monitoring approaches are being embraced as time and resources saving tactics, and the type of health monitoring we are employing at EDEN ISS will contribute to the sophisticated tool box being employed in this expanding technology.

	ISS Synthetic Experiment Following work done with James Goodman of HySpeed Computing. Demonstration of Onboard Image-Based Plant Health Monitoring using ENVI in Space Developing automated systems for managing plant cultivation are important for the success of long-duration space missions. Towards that objective, we are utilizing the compute capabilities of the ENVI in Space mission to demonstrate a methodology for image-based monitoring of plant health on the International Space Station (ISS). ENVI in Space has deployed a full enterprise version of the industry leading image processing and analysis software ENVI® on the Hewlett Packard Enterprise (HPE) Spaceborne Computer-2 (SBC-2), which is located on the ISS. SBC-2 is the first commercial off-the-shelf high performance edge computing system with artificial intelligence (AI) capabilities built for in-space data processing. Through this system, ENVI is being used to illustrate the advantages of onboard computing for an array of imaging applications, from Earth observation to space exploration. Here we are utilizing multispectral timeseries of plant images from the EDEN-ISS project and other controlled laboratory experiments to showcase image analysis capabilities for automated monitoring of plant health. This includes a full data processing workflow, from data ingest and pre-processing to spectral computations and statistical analysis. Results are generated in a user-friendly format to simplify interpretation and streamline future implementation as an automated plant management tool. Multispectral and AI integration Work done in conjunction with Dr. Sanjeev Koppal and Ms. Anna Hampton.
Task Progress:	Convolutional Neural Networks (CNNs) have shown exceptional performance in exploiting spatial relationships in image data. Given that the key indications of saline induced stress in our sample is loss of turgor pressure and leaf shrinkage, we have chosen to employ a CNN to extract feature maps. Following the convolutional layers, the features are linearized and passed through fully connected layers with the final layer producing the predicted time since stress. As far as image preprocessing, images were broken down into the four quadrants that correspond to the 4 plates. Labels for each quadrant image were generated based on time since stress exposure. The images pre-stress (or within the buffer times) and the control quadrant images are assigned a label of 0. A buffer region post application of saline is being used to ensure we have sufficient time for stress to set in before we assign a label indicating stress is present. We are experimenting with two different types of labeling, one with basic time since stress and one with exponentiated time since stress. The exponentiated time since stress is being tested because the network may learn better if there is a larger distinction between stress and non-stress images. Project Summary (Ed., November 27, 2022) The spectral imaging project has moved forward the understanding of a deployable Plant Health Monitoring system for future off-Earth applications. The ability to build, calibrate and deploy an imager system, that cannot only detect early signs of stress in a plant, but that can also facilitate and early response will be important as enclosed life support systems become commonplace in off-Earth environments. The addition of potential on board processing and AI image analysis will continue to improve the future of plant heath detection systems in the future.
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